

Fairness for Dating Apps: Mitigating Discrimination by Preventing Popularity Bias

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Dating apps are important intermediaries in people finding romance, yet dating algorithms are likely to discriminate on basis of ethnicity. Dating apps cannot sufficiently measure or mitigate discrimination, as the GDPR prevents dating apps from accessing or processing ethnicity. In this work, we show that popularity bias can be a driver for discrimination, and that mitigating popularity bias can result in reduced algorithmic discrimination in content-based recommendation. Interventions on popularity bias appear more effective in content-based recommendation, with an appropriately chosen definition of popularity bias, and targeting indirect discrimination, rather than direct discrimination.

Keywords: Discrimination, Algorithmic Fairness, Dating app, Dating algorithm

Reference Format:

Tim de Jonge and Djoerd Hiemstra. 2026. Fairness for Dating Apps: Mitigating Discrimination by Preventing Popularity Bias. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26)*. Proceedings of Machine Learning Research, 21 pages.

1 Introduction

A recent case brought to the Netherlands Institute for Human Rights (henceforth: the Institute) indicates that dating apps could indirectly discriminate on the basis of ethnicity [14, 16].

When Breeze, a Dutch dating app, surveyed their users, some of their users complained that Breeze was not showing them enough ethnic variety: most of the profiles they saw belonged to white-skinned users.¹ Breeze suspected that their algorithm discriminated against profiles of dark-skinned users. Breeze found from literature that the average Dutch person is more likely to like a profile from a light-skinned user than a profile from a dark-skinned user, and that matching algorithms are prone to give popular profiles more attention than they deserve.

They combined these two claims: if a profile of a dark-skinned user is less likely to be popular, and unpopular profiles are likely to get less attention than they deserve, it is likely that the profiles of dark-skinned users are getting less attention than they deserve. Breeze suspected that this constitutes discrimination on grounds of ethnicity, and asked the Institute whether they were allowed to address this discrimination.

¹A quick remark on terminology: the decision remarked that Breeze might discriminate against people *with darker skin or non-Dutch heritage*. For the sake of brevity, we refer to the users as light-skinned or dark-skinned, and to the attribute as ethnicity.

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ECAF'26, September 02–September 04, 2026, Ghent, BE

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The Institute accepted this reasoning as it stands, although Breeze did not provide statistical evidence for either claim in their app. In particular, the Institute highlighted that the law allows for objective justification for this discrimination. Since Breeze did not provide a justification, the Institute proceeded under the assumption that Breeze discriminates illegally. De Jonge and Borgesius [16] discuss the nuances of the possibility for justification in this exact case, while Tobler [41] provide an excellent overview of the legal intricacies of indirect discrimination in general. The Institute responded that if Breeze discriminates illegally, they are obligated to address this discrimination. The Institute did not provide any advice on *how* Breeze should be addressing this discrimination, only *that* Breeze should address the discrimination.

It is not easy to address discrimination on the basis of ethnicity under European law. Most of the literature on mitigating discrimination is written under the assumption that developers have access to ethnicity data. In Europe, the GDPR effectively forbids the use of ethnicity data. This prevents some discrimination outright as platforms and users do not have the option to filter directly on ethnicity, but it makes it illegal to even measure the level of discrimination in a system. Van Bekkum and Borgesius [43] outline the debate around allowing the use of ethnicity data for the prevention of discrimination.

In this work, we investigate the possibility of addressing discrimination by mitigating popularity bias. Breeze reasoned that popularity bias was one of the main causes of discrimination in their system, leading us to suspect that removing popularity bias would reduce discrimination. As such, our central research question is:

Can a dating app mitigate discrimination in their algorithm by reducing popularity bias?

To answer this question, we investigate how the relation between discrimination and popularity bias changes when users' romantic preferences correlate more with ethnicity. We distinguish direct user discrimination from indirect user discrimination. A user discriminates directly on the basis of ethnicity when the user treats profiles differently for the reason of ethnicity in and of itself, and a user discriminates indirectly when the user prefers profiles with characteristics that correlate with ethnicity; as such, our sub-questions are:

RQ1: Can a dating app mitigate discrimination their algorithm caused by *direct user discrimination* in their algorithm by reducing popularity bias?

RQ2: Can a dating app mitigate discrimination their algorithm caused by *indirect user discrimination* in their algorithm by reducing popularity bias?

We return to the distinction between direct and indirect user preferences in more conceptual detail in section 2.3, and in more technical detail in section 3.2.

2 Related Work

This paper investigates the connection between popularity bias and discrimination in dating algorithms. Although popularity bias and discrimination are closely related, and dating algorithms are a natural field of study for these concepts, there is little previous literature in the intersection of these concepts. We discuss each concept in turn, highlighting work on the intersections where it exists.

First, we introduce the concept of *recommender systems*. A dating algorithm is a particular case of a recommender system, and popularity bias and discrimination are well-studied in the field of recommender systems. Recommender systems are incredibly common around the modern web, and as such, there is a great deal of literature on recommender systems. Where literature on dating apps is limited, we will draw inspiration from the literature on recommender systems instead.

A recommender system is an algorithm that provides a user with a set of recommended items that the system predicts that the user will like. [5, 10, 34]. Here, we need to interpret "items" in a broad sense. A canonical example of a recommender system exists inside a web shop. Web shops typically contain a section labeled "Users like you also liked..." or "If you liked X, you may also like..." The web shop recommends a user those products that user might like. Another example exists in music streaming services: after a user plays a song, the music platform might continue to play songs which the user did not explicitly ask for, but the platform judges the user might be interested in. In particular, dating algorithms are examples of recommender systems: users want to see profiles of people they might be interested in, and dating algorithms recommend those profiles to the users.

A dating app is a program where users can attempt to find romance. On a dating app, users create a profile, typically involving the users' pictures, primary demographics, romantic preferences and whatever other information the user wants to volunteer. Then, the app will offer the user the profiles of other users, matching the user's romantic preferences. The user can then express interest in the profile of an other user, typically called a *like*. If two users like each other, they will *match*, allowing the users some further communication; in the case of Tinder, either user can message the other - in the case of Bumble, the woman can message the man - in the case of Breeze, users are guided towards an interface where they can set up a curated date.

By and large, the online dating industry is unwilling to share much details about the way they match users, although there are some notable exceptions [22, 39]. Typically, a dating app will use an algorithm to calculate which profiles to show a particular user, using data including users' interests, activity on the app, and previous likes. In 2013, OKCupid's lead data scientist recorded a short TED-ED film detailing the functioning of their algorithm. OKCupid allows users to fill in answers and preferences on a vast array of questions, and then calculates a *match percentage*, describing how well users' profiles and preferences line up with each other [30, 36]. Grindr claims their service uses a very straightforward algorithm, as they claim Grindr "displays those who were online recently, and applies the searching user's filters [...] sorted by distance. Sometimes a little randomness is thrown in to keep results fresh. That's it." [46]. Beyond these cases, we only have very broad information about the functioning of the matching algorithms of dating apps.

2.1 Popularity Bias

A recommender system suffers from *popularity bias* when it recommends popular items even more frequently than their popularity would warrant [3, 8]. Klimashevskaja et al. [25] synthesize existing definitions into "A recommender system faces issues of popularity bias when the recommendations provided by the system focus on popular items to the extent that they limit the value of the system or create harm for some of the involved stakeholders." This popularity bias tends to arise due to a self-reinforcing effect also known as the Matthew effect [20, 27, 33]. Recommender systems frequently incorporate the preferences of users into their recommendations: if you resemble users that like a particular item, you might also like that particular item. This could lead to a self-reinforcing loop: since users like an item, the system is likelier to recommend it to more people. Since the system is eager to recommend the item, more users will like it. If left unchecked, this reinforcement loop could spiral out of control, where popular items remain popular due to their popularity, even when newer, better options appear. We will return to the operationalization of popularity bias in section 4.2.

There is little work directly addressing popularity bias within the realm of dating apps. In a seminal paper, Pizzato et al. [34] describe the differences between a typical recommender system (e.g. a search engine), and

a dating app. In this paper, they note how people have limited availability, and as such, that a dating algorithm should not recommend a profile too many times to prevent the schedules of the users overflowing. Paraphrasing: a dating algorithm should avoid popularity bias. In a follow-up paper, they describe an initial intervention to mitigate popularity bias in a dating algorithm [35], which we will revisit in section 3.4.

Since these initial offerings, research on dating algorithms has only tangentially interacted with popularity bias. Several papers produce recommender systems following fairness desiderata unrelated to popularity bias, which limit the amount of popularity bias in the system as a side-effect [42, 47]. These papers do not engage with other literature regarding popularity bias: they target their own fairness desiderata, which indirectly restrict the number of recommendations a popular profile can get.

Celdir et al. [12] do directly mention popularity bias, but in the context of dating app policy. Some dating apps rely on a premium, paid version for their income (e.g. Tinder Gold [40], Bumble Premium [9]). The dating app prioritize displaying the profiles of paying users, and paying users would have a limited ‘priority queue’ to more directly address users they are interested in. The dating app might then be incentivized to initially direct users towards less attractive profiles, so that they pay for the premium version, to gain access to the more attractive profiles [12]. They conclude that dating apps don’t significantly benefit from optimizing for premium subscriptions, compared to optimizing for matches in the system.

There is some prior research suggesting a link between popularity bias and fairness. Ekstrand et al. [19] suggest that popularity bias can, but does not always, lead to differences in model performance between different groups. If one group systematically gets better recommendations than another, this can be interpreted as discrimination, or unfairness. Abdollahpouri et al. [2] *define* a system as unfair if model performance differs for different users. They find that popularity bias can indeed lead to different model performance, and thus, unfairness. Ionescu et al. [21] investigate the connection between popularity bias and discrimination within the domain of content creation, and found that a reduction of popularity bias could also reduce discrimination, albeit insufficiently to entirely eliminate it.

2.2 Discrimination in Dating Algorithms

We can distinguish two major strains of relevant literature to discrimination in dating algorithms: the fair recommender system literature, and the literature addressing dating apps as technological systems outside of the algorithm. First, there are several works on sexual racism on dating apps, through different demographics and localities. This literature on sexual racism focuses primarily on discrimination by users, and primarily in the context of proximity-based dating apps, which do not feature extensive matching algorithms (e.g. [15, 37, 44]). As such, this strain of work complements our approach to form a full picture of discrimination in dating apps.

Second, there is technical literature that could reasonably be applied to discrimination under the umbrella-term of *fairness* in recommender systems [17, 45]. This literature contains a broad range of measurements and interventions on difference in outcome between groups. As canonical examples, Zehlike et al. [50] suggest an interleaving approach, where the system ranks majority and minority candidates separately, and interleaves these candidates to form a ranking with the desired balancing between majority and minority candidates. Morik et al. [28] explicitly calculate the difference in outcome between majority and minority, and boost the scores of minority items in proportion to this difference, resulting in balance of outcome, over time. Oosterhuis [31] further shows that a certain

class of commonly used recommender systems could control for difference of outcome between groups, as long as this difference adheres to minimal constraints.

These works require access to group membership to measure and mitigate bias. The GDPR forbids collection or inference of ethnicity, meaning these works cannot be applied to discrimination on grounds of ethnicity where the GDPR applies.

Last, the Netherlands Institute for Human Rights conceptualize discrimination as: it is discriminatory if an algorithm shows profiles of people of a given ethnicity less frequently than profiles of people of a different ethnicity. The Institute held that this was the case regardless of the preferences of people on the app - the algorithm was not allowed to amplify or reproduce these preferences. In this paper, we maintain this definition of discrimination given by the Institute. For a further discussion on how to measure algorithmic discrimination in dating, we refer to [16].

2.3 Racial Preferences in Partner Choice

Modeling attraction is a difficult endeavor, as much of it depends on chance. In order to be able to run a simulation, we base ourselves on social theory - the real world contains a great deal more noise than we present in our model, but the following theory informs our choices.

We begin with assortative dating as theory of attraction, which claims that individuals match up with people with similar traits (e.g., education, intelligence, health, religion) [6, 7].

Additionally, literature points out that sexual racism expresses itself in several different ways. As a first split, Chiswick and Houseworth [13] note that individual preferences and opportunities for contact are the two primary determinants of interethnic marriage, with other factors influencing these two. This theory implies that dating apps could stimulate interethnic marriage, as dating apps could remove some of the boundaries that currently exist for social contact between people of different ethnicity.

Lucassen and Laarman [26] further differentiate "individual preferences" into direct preference for a phenotype (external expression of ethnicity/race) and preferences that do not directly relate to ethnicity, but are correlated, such as religion, or moral values. They note that the relative strengths of these preferences can differ: Europeans place less importance on phenotype when compared to US citizens.

In the following text, we call users preferring users with their own ethnic or racial phenotype *direct user discrimination*. When users have a preference that correlates with ethnicity, but is not caused by ethnic or racial phenotype, we call this *indirect user discrimination*.

3 Simulation Setup

In the following section, we describe the details of our simulation. The entire code for simulation and evaluation is available on <https://github.com/TimdeJonge/fairdating>.

3.1 Summary

We simulate a dating app with 1000 users: 900 of these users will be part of the ethnic majority, the remaining 100 will form the minority. In our experiments we will vary the levels of indirect and direct user discrimination on the basis of ethnicity, as defined in section 2.3. We speak of direct user discrimination when users make judgments directly on the ethnicity on the profile they see: if the profile does not match their own ethnicity, they

deem the profile less attractive. We speak of indirect user discrimination when users judge profiles in a way that correlates with ethnicity, even though they do not judge on phenotype directly. In this case, the majority and minority population have different characteristics and preferences. For example, the majority might consist of taller people who want to date other taller people, whereas the minority predominantly consists of smaller people who want to date other smaller people. While users only judge on height, not ethnicity, their preferences will correlate with ethnicity.

In our simulation, we use two commonly used baselines: RECON [34] and RCF [48]. We evaluate the effectiveness of an intervention that mitigates popularity bias, that we will refer to as *PB* [35]. *PB* limits the number of times an algorithm can recommend a profile, meaning no profile can become too popular. *PB* further forces all recommendations to be mutual: if the algorithm shows profile A to user B, the algorithm must also show profile B to user A. Users should then all receive approximately the same number of recommendations, resulting in a more balanced model performance.

We evaluate these models on 3 characteristics: performance, popularity bias, and discrimination.

For performance, we assess how many recommendations result in likes, expressed as percentage.

For popularity bias, we use two of the metrics most commonly used to assess popularity bias [25]. We use the Gini coefficient as measure of skew in the number of times the algorithm recommends profiles [4, 18]. The more the algorithm displays the most popular profiles in contrast to the last popular profiles, the higher the popularity skew, and the higher the Gini coefficient. An algorithm that only displays one profile has a Gini coefficient of 1; an algorithm that displays all profiles equally has a Gini coefficient of 0.

Additionally, we assess popularity bias through Average Rank Popularity (ARP), which calculates how popular the average recommended profile is [1, 49]. In effect, this estimates whether the algorithm favors profiles that many users like. If a model displays profiles of attractive users more frequently, ARP is higher than 1, and if the model displays profiles of less attractive users more frequently, ARP is lower than 1.

For discrimination, we calculate Demographic Parity (DP), as a ratio, which calculates how frequently the algorithm displays the profile of an average majority user compared to the profile of the average minority user. If the algorithm displays profile of majority users more frequently, DP is higher than 1, and if the model displays profiles of minority users more frequently, DP is lower than 1.

3.2 World

Table 1 details the notation used in subsequent sections.

We define 1000 profiles x_i , with attributes X_i^k being the k^{th} attribute for user i . There are 3 types of attributes:

First, x_i^0 is ethnicity, or e_i . The simulated population is made up of an ethnic majority making up 90% of the population ($e_i = 0$) and an ethnic minority making up the remaining 10% ($e_i = 1$). Ethnicity is never visible to any of the algorithms, but ethnicity is visible to all users. Then, X contains several attributes unrelated to ethnicity. For $k \in [1, 2, 3, 4]$, $X_i^k \sim U(0, 1)$, or the first four attributes are drawn uniformly from $[0, 1]$. Finally, X contains several attributes correlated with ethnicity. For these attributes, we draw from normal distributions, where the norm for the ethnic minority is offset by λ from the norm for the majority: $k \in [5, 6, 7]$, $X_i^k \sim \mathcal{N}(\lambda e_i; 1)$. We restrict these values to the interval $[-1, 1]$. We use λ to control the users' level of indirect user discrimination: if λ increases, the profiles of majority and minority users will be further apart, meaning users prefer profiles from their own ethnicity more.

Symbol	Description
x_i^k	Attribute k for user i
x_i^0, e_i	Ethnicity
λ	Indirect discrimination parameter
γ	Direct discrimination parameter
L_i	users that i Liked, $\{j : i \text{ liked } j\}$
W_i	users that i Was liked by, $\{j : j \text{ liked } i\}$
δ_i	x_i transformed into Dummy variables
S_{ij}	Model calculated Score of profile j to user i
R_i^k	k^{th} Recommendation for i
R_i	Total list of Recommendations for i
D_i	Number of times the model Displays i 's profile
$\phi(i)$	Popularity: here, number of users that liked i 's profile ($\ W_i\ $)

Table 1. Summary of notation used in this work.

In our simulation, some users prefer profiles of users matching their own ethnicity. Each user has a value r_i that determines whether they discriminate directly: a user prefers profiles matching their own ethnicity when $r_i = 1$, otherwise $r_i = 0$. r_i has a 50% chance to be 1, and a 50% chance to be 0. r_i is not visible to users, or to the matching algorithm. γ controls the intensity of direct user discrimination: a high γ means ethnicity is an important factor in users' attractions.

Users are attracted to each other when the Euclidean distance between their profiles is sufficiently low. Combining this with the previously mentioned direct user discrimination, user i is attracted to j when $\|x_i - x_j\|(1 + r_i\gamma) < 0.9$ [38]. A user will like a profile if they see the profile and they are attracted to the profile. We denote by L_i the set of profiles that i liked, i.e. $L_i = \{j : i \text{ liked } j\}$.

As a starting point for our recommendation algorithm each user sees a random sample of 10% of the population. This is a fairly robust sample of the population, ensuring that even unpopular profiles typically have some positive interactions.

3.3 From Users to Scores

3.3.1 RECON

RECON is one of the first algorithms aimed at recommendation for dating [34]. Still, it is a commonly used baseline for new dating algorithms [32, 47]. In short, if RECON evaluates whether to recommend profile j to user i , RECON checks all profiles that i has liked before. For each attribute that j has, it sees how common that attribute is among people i has liked. RECON then averages these frequencies, and that is the score RECON gives profile j for user i .

RECON operates only on discrete attributes, so we must start by making our attributes discrete. For ease of calculation, we work with the dummy variables of these discrete variables; δ_i , where δ_i contains the same information as x_i , but now as dummy variables. Then, we can define the preferences of i as the number of profiles that i has liked with a particular attribute: $P_i = \sum_{p \in L_i} \delta_p$. Note that P_i is a feature vector of the same length as δ_i .

Then we can denote RECON as

$$\text{RECON}(x_i, x_j) = \frac{P_i \delta_j}{\|A\| \|L_i\|}$$

Where $\|A\|$ is the number of attributes in X , and $\|L_i\|$ is the size of the set L_i . This definition of RECON is equivalent to the original definition, but easier to manipulate and update in vector-typed languages.

Let us look at a quick numerical example to explore this formula.

Characteristic	Profiles i has seen	Profile i has liked
Likes Singing	5	5
Does not like Singing	95	10

Table 2. Data for a hypothetical user i , where the dating app only considers singing as attribute

To ease calculations, we are only looking at one characteristic here: whether a user likes singing or not. In table 2, we have aggregate numbers for a hypothetical user i . Suppose RECON has to score two new profiles for user i : the profile of j , who does not like singing, and the profile of k , who does. RECON sees that i has liked 15 total people, of which 10 did not like singing. RECON assigns j a score of $\frac{10}{15} = \frac{2}{3}$. RECON assigns k a score of $\frac{5}{15} = \frac{1}{3}$. RECON thus scores the profile of j higher than the profile of k , as j resembles the users i has liked more closely.

3.3.2 RECON*

In exploratory analysis, we found that RECON is biased towards more common traits, regardless of user preference. RECON has the potential to mistake a common trait for a desirable trait. Again, look at table 2. RECON recommended j over k , since j most resembles users i has liked. However, the rows of table 2 tell a different picture: i liked every single profile of people that liked singing, yet only $\frac{10}{95}$ of the profiles of people that did not like singing. RECON recommends the more common trait, even though the user appears more attracted to the less common trait.

To better align the recommender to user preferences, we propose an adapted version of RECON, which we name RECON*. RECON calculates how similar a profile is to profiles the user has liked before. RECON* instead calculates what percentage of the time a user liked a profile with that particular trait. This aligns better with the goal of predicting whether a user would like a particular, previously unseen, profile. We formulate RECON* as follows:

$$\text{RECON}^*(x_i, x_j) = \frac{1}{\|A\|} \frac{P_i \delta_j}{\sum_{n \in N_i} \delta_n}$$

where N_i denotes the set of users i has interacted with, positively or negatively. To return to the example data of table 2, with user j that does not like singing, and user k that likes singing, RECON* assigns to j a score of $\frac{10}{95} \approx 0.1$ and RECON* assigns to user k a score of $\frac{5}{5} = 1$. RECON* scores the profile of k higher than the profile of j , as i is more likely to like k .

3.3.3 RCF

We further use the CF2 model described by Xia et al. [48], which has later been dubbed RCF and is commonly used as baseline for Collaborative Filtering systems in dating apps [24, 29, 32]. If we denote those users who liked

j as W_j , we can denote RCF as

$$\text{RCF}(x_i, x_j) = \frac{1}{\|W_j\|} \sum_{w \in W_j} \frac{\|L_i \cup L_w\|}{\|L_i \cap L_w\|}$$

In words: RCF estimates whether i will be interested in the profile of j . To estimate this, RCF looks at all the people w that have already liked j . If w and i have liked a lot of the same people, it is likelier that w and i have similar taste. Since w likes j by definition, that makes it likelier that i also likes j . Or: The more similar i is to the people that liked j , the more likely it is that i also likes j .

3.4 From Score to Ranking

RECON, RECON*, and RCF are just scoring algorithms; we need another step to get from the scores to a ranking. Suppose user i wants to receive a list of recommendations: R_i . In this section, we take RECON as example; the case for RECON* or RCF works analogously. First, RECON estimates how interested i would be in the profiles of other users with scores $\text{RECON}(x_i, x_j), \forall x_j \in X$. However, a dating app is an inherently two-sided system. In order to provide i with matches, we must also recommend profiles of users that might be interested in i [34]. As such, we also need to calculate all scores $\text{RECON}(x_j, x_i), \forall x_j \in X$, and then combine the scores for i to j and j to i into one score. Most common is to use the harmonic mean of scores:[29]

$$S_{ij} = \frac{2}{\text{RECON}(x_i, x_j)^{-1} + \text{RECON}(x_j, x_i)^{-1}}$$

Then, we can form a list of recommendations by suggesting to each user the profiles with the highest scores that the user hasn't seen yet, i.e. the k^{th} recommendation for user i is given by

$$R_i^k = \max_{x_j \in X/R_i^{k-1}} S_{ij}$$

This way of ranking does not control for popularity bias, and as such, will probably display some skew in the distribution of popularity. Pizzato and Silvestrini [35] suggests two restrictions to a recommender system to limit popularity bias:

- When you recommend user i to user j , you must also recommend user j to user i .
- Limit the amount of profiles a user can see to a maximum of m .

These two restrictions together guarantee that the system cannot recommend a profile to more than m users. This ensures that the system does not recommend popular profiles too frequently, and thus, that the gap between the most popular and the least popular items cannot be too wide.

Pizzato and Silvestrini [35] propose a greedy algorithm to satisfy these two restrictions, which we refer to as *PB*. *PB* calculates all scores between users. At every step in the algorithm, *PB* takes the highest score between a pair of users i and j , and fixes that pair as a recommendation: *PB* recommends i to j , and j to i . If this means that i now has m recommendations, *PB* sets the scores of all pairs involving i to 0, so i does not receive further recommendations; similarly for j . Then, *PB* sets the score between i and j to 0, as *PB* has already recommended i and j to each other, so i and j need not be recommended to each other again. *PB* then repeats this iteration until there are no more scores above 0, or until all users have received m recommendations.

It is possible that this algorithm terminates before all users have m recommendations, when users are too unpopular; in this implementation, we have elected to leave these slots blank. A different option would be to fill

these slots at random with other users who have received less than m recommendations. As phrased, this algorithm recommends all users simultaneously; this is not strictly necessary. Asynchronously, every user would get space allotted for m recommendations. Once a user visits the site, if the system has m recommendations ready, the system shows the user the m recommendations. Otherwise, the system fills the space with other profiles that still have open space, in descending order of score. There would then be a cool-down period before the user can visit their new list of m recommendations.

4 Evaluation

In this work, we evaluate based on three criteria; how good the algorithm is at matching different users, whether the algorithm has popularity bias, and whether the algorithm discriminates.

4.1 Accuracy

For system accuracy, we focus on like rate. The like rate is given by the percentage of recommendations that results in likes, i.e. Like rate = $\frac{\sum_{x_i \in X} \|L_i\|}{\sum_{x_i \in X} \|R_i\|}$. This implementation of like rate aligns with precision, in traditional information retrieval terms.

Usually, it would be more appropriate in a dating system to focus on match rate. However, in our simulation the attraction function is mostly symmetrical, meaning that A likes B, B typically also likes A. Then, PB has as side effect that the system recommends A to B if and only if the system recommends B to A. This means that most likes in a system with PB also result in matches. As such, evaluating on match rate means PB gets valued much higher than appropriate.

4.2 Popularity Bias

In this work, we align the judgment of popularity bias with the definition of discrimination given by the Institute. The Institute held that a dating algorithm discriminates when it shows the average profile of a dark-skinned user less frequently than the average profile of a light-skinned user. This motivates the choice to name all differentiation on the basis of popularity as popularity bias, i.e. no differentiation on the basis of popularity is warranted. We can conceptualize the popularity bias of an algorithm as the inequality of the amount of times the algorithm displays profiles. The Gini coefficient of displays D_i can be calculated as

$$\text{Gini}(D) = \frac{1}{n^2 \bar{D}} \sum_{i=1}^n \sum_{j=1}^n \|D_i - D_j\|$$

where $\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i$ [18].

Abdollahpouri et al. [1] define the Average Rank Popularity (ARP) as:

$$\text{ARP} = \frac{1}{n} \sum_{x_i \in X} \frac{\sum_{r \in R_i} \phi(r)}{\|R_i\|}$$

where $\phi(r)$ is a measurement of the popularity of a given user. In this work, we define $\phi(x_i)$ as the number of times x_i has been liked in the initial condition, after we showed each profile to a randomly sampled 10% of the population.

4.3 Discrimination

We calculate discrimination according to Demographic Parity (DP): $DP = \frac{D^+}{D^-}$. In this equation, we take D^+ to be the average number of times the system has displayed the profile of a user of the privileged group, where D^- is the equivalent for the group discriminated against.

We run two experiments: First, we vary the value of λ , while keeping γ stationary at $\gamma = 0.1$. This investigates the effects of distributional difference between the majority and minority group, and as such, the effects of indirect user discrimination. Second, we vary the value of γ , while keeping λ stationary at $\lambda = 0.2$. This investigates the effect of direct user discrimination.

All experiments are run 30 times with random seeds that have not been seen before the making of these graphics; all values reported are averages over these 30 runs.

5 Results

Figures and tables detailing the full results can be found in the appendix.

In figure 1 we see the results of manipulating λ , or the extent to which the majority and the minority distributions differ. If the majority and minority are more distinct, majority and minority users will be less attracted to each other. Since users are attracted to users like themselves, this happens even without the users directly considering ethnicity.

In the top-left, top-right, and bottom-left parts of figure 1, we see that varying λ does not impact model accuracy, popularity bias in any significant way. In the bottom-right of figure 1, we see that as λ increases, RECON increasingly favors the minority, as RECON* favors the majority. As λ increases, users in this model are less attracted to users with different ethnicity, and so, users have stronger racial preferences. We can see that RECON and RECON* pick up these preferences, albeit in different directions.

In all cases, the recommender algorithms without PB are marked with crosses, while the recommender algorithms with PB are marked in the same color and line-style, but with dots. RECON is marked in dash-dotted red(-.-), RECON* in dotted green, RCF in dashed blue, and a fully random baseline in solid black. As we can see in figure 1d, RECON, without PB, discriminates in favor of the minority. We can see in figure 1b that RECON does have popularity skew: attention is unequally distributed across users. In figure 1c, we see that ARP is below 1; meaning that RECON favors users that are deemed less attractive than average. In figure 1a, we can see that PB slightly boosts RECONs accuracy. Across all levels of indirect user discrimination, RECON* scores higher like-rate than RECON. Adding PB appears to level this out - RECON + PB and RECON* + PB are closer together than just RECON and RECON*. In figure 1b and 1c we see that PB greatly reduces the level of popularity bias. After PB, RECON and RECON* both perform better than random, whereas before they performed much worse. Figure 1d shows that RECON has discrimination in favor of the minority, and RECON* discriminates in favor of the majority, increasingly as indirect user discrimination also increases. PB mitigates discrimination in both these cases to noise levels.

For RCF, PB is less effective. In figure 1b, we see that PB still reduces popularity skew to just below the level of randomness, and figure 1c shows that PB reduces attractiveness bias slightly. PB has almost no effect on discrimination for RCF in this setup. Overall, we can see that the models' accuracy or popularity bias do not change as a result of indirect user discrimination, but discrimination does change for all models, particularly for RECON(*).

Again, we see many of the same patterns as in the indirect user discrimination setup. In figure 2a, we see that for all models, like rate drops as direct user discrimination increases, including for the fully random model. Again, RECON* outscores RECON significantly, but RECON*+PB only slightly outscores RECON. RCF achieves higher like rate than the RECON models, and RCF+PB performs best on like rate. In figure 2b, we see that PB is effective at reducing popularity skew for RECON models. If we contrast figure 1b with figure 2b, we see that, as user discrimination increases, more popularity skew remains in the system, across all models. Figure 2c shows that PB is also effective at reducing attractiveness bias. Again, RECON initially benefits those deemed less attractive, while RECON* heavily benefits those deemed more attractive. The PB intervention almost completely levels these benefits. PB also mitigates the attractiveness bias in RCF, but to lesser extent; even after PB, attractiveness bias remains. There is no discernible pattern in figure 2d.

6 Discussion and Limitations

We have investigated whether PB ([35], detailed in section 3.4) results in reduced discrimination, across 3 algorithms and 2 conditions. Figure 1b shows that both RECON and RECON* have popularity skew above the baseline. Figure 1c reveals RECON, in this circumstance, is biased towards those less popular, whereas RECON* is heavily biased towards those more popular. Figure 1d then adds that PB reduces discrimination to approximately noise levels for RECON and RECON* for all choices of parameter. RECON and RECON* have a similar Gini-coefficient, but opposite directions of discrimination, or attractiveness bias. Despite both being commonly used measurements of popularity bias, Gini and ARP measure different constructs, and should not be used interchangeably.

PB achieves much less reduction of popularity bias for RCF, compared to RECON(*). This can be understood by contrasting the functioning of these algorithms. RCF relies on the behavior of other users to recommend profiles. The more people have liked a profile, the higher RCF scores that profile. Even though our simulations initiate users with 100 interactions with other profiles, some profiles do not get enough likes to generate 10 recommendations.

PB aims to adapt the ranker so the system shows each profile 10 times. If all profiles were shown exactly 10 times, that would lead to 0 popularity skew. When applied to RCF, PB runs out of profiles to recommend. PB generates recommendations in order of highest score - the most popular users have the highest scores in RCF, so PB prioritizes the recommendations of the most popular users. As such, when PB starts to find recommendations for an unpopular user, the system will already have ranked the many, more popular users the maximum amount of times. The system can then not recommend these popular users to the unpopular user. Statistically, it is unlikely that unpopular users have many likes in common. As a result, PB might even lower the number of times the system shows the profile of an unpopular user, as the system shows more profiles of popular users to other popular users. Other interventions on popularity bias might achieve greater effect on collaborative-filtering recommender systems.

6.1 Limitations of the design

As this work is exploratory in nature, the system design has some clear limitations that could be addressed in future research.

The starting situation for this experiment is more balanced than is typically the case in recommendation research. The system shows each profile approximately 100 times, and differences between users are not too stark in terms of attractiveness. In the real world, popularity bias can have a self-reinforcing dimension: an algorithm will display the

most popular users most, which leads to these users getting more likes, which results in the algorithm displaying their profiles more. In this experiment, we are working from a relatively balanced starting point, and only working through one temporal step. Further research could investigate the effects of multiple time steps on discrimination and popularity bias in a collaborative filtering model.

In the "Direct Discrimination" condition, results are inconclusive. The current model outcomes have too much variance to adequately assess whether anything changed across conditions. This hints at a further potential difficulty in assessing discrimination in these systems; even a decently sized system of 1000 users, across 30 runs is insufficient. This means that assessing discrimination in actual dating applications needs to be done with care, to ensure that any statistical result is not just a fluke.

RECON appears to be a fairly weak baseline. As we explain in section 3.3.2, RECON recommends profiles that resemble profiles you have liked before, which is not the ideal target. Across all conditions, RECON* outperforms RECON. In figure 1b we saw that RECON* recommends popular users more often, which is one part of the performance improvement. However, even after PB, RECON* performance remains slightly higher, meaning RECON* does not just exploit popularity bias. Interestingly, RECON+PB also outperforms RECON consistently. Xia et al. [47] have previously increased RECON performance with an out-of-the-box fairness intervention, both on fairness and on match rate. We hope that RECON* can be an adequate start towards research for a stronger dating algorithm baseline.

This work holds no pretense at capturing all relevant social complexities. The model disregards gender completely, while gender influences user ethnic preferences, user behavior, and any number of other factors. We model ethnicity as a binary characteristic to align with the legal definition: we contrast a group we suspect the system discriminates against, with a baseline. In the real world, ethnicity is a complex, multi-dimensional, continuous construct, that may not be measurable at all [11, 23]. Interdisciplinary cooperation is needed to apply this work to real-world scenarios, to ensure that the appropriate legal, ethical, social, and even economic perspectives are taken into account.

7 Conclusion

The GDPR outlaws all processing of ethnicity data, making it difficult to measure or prevent discrimination in digital systems. In this work, we show that popularity bias can be a useful proxy for discrimination. We investigate the effectiveness of an intervention, here called, PB, that places a maximum on the number of times the system can show a particular profile in a given time frame. Since PB limits the range of possible outcomes, PB limits the possible difference in treatment of the ethnic majority and the ethnic minority. With that, PB also limits discrimination as conceptualized in this paper.

We tested two conditions in our simulation: One where the users' racial preference were a result of differences in attributes between majority and minority, and one where users' preferences were directly influenced by ethnicity. In the 'indirect' condition, we find that PB consistently reduces differential treatment on the basis of ethnicity for a content-based dating algorithm. PB does not appear to not have much effect for a collaborative-filtering algorithm, as the greedy nature of PB results in poor recommendations for unpopular users. In the 'direct' condition, we have not been able to find stable results.

The relation between popularity bias and discrimination also depends on the measurement of popularity bias: we find that the Gini-coefficient measures to what extent the system distributes attention equally, whereas Average Rank Popularity measures to what extent the system attention is distributed to profiles that are already popular.

When the system skews attention in favor of the unpopular, this means that the Gini-coefficient and Average Rank Popularity provide opposite readings. Since discrimination is about who the system distributes attention to, Average Rank Popularity is a more appropriate measurement to use as proxy for discrimination.

In conclusion, this paper demonstrates the difficulties in measuring discrimination in digital systems. This paper confirms the hypothesis that popularity bias can be a factor in discrimination, and that mitigating popularity bias could mitigate indirect discrimination in a model. However, this paper casts doubt on the claim that individual fairness always results in group fairness.

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